

Compact Representations of Human Interception Movements

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Working draft. Red brackets identify questions for Jason.

1 Study objective and experimental data

We investigated whether a compact numerical fingerprint could capture individual differences in interception movements and reproduce a participant's movement distributions. The principal setting used three latent values, with eight providing a higher-capacity comparison. Neural encoder-decoder models were compared with spline-based dimensionality reduction and regression on reconstruction, feature distributions and individual information. These comparisons inform model selection for an interactive dashboard.

1.1 Experimental task

The recordings were provided by Jason Friedman's laboratory. Participants moved their finger from a starting region toward a fixed interception region. A circular target appeared on either side of the display and subsequently moved horizontally toward the interception region. Participants were instructed to reach that region while the moving target was inside it. Successful performance therefore required coordinating finger arrival with target arrival (Figure 1).

The target appeared at one of six starting locations: three on the left and three on the right. Each starting-position category was associated with a range of speeds, from which a continuous target speed was randomly selected for each trial. Starting position and speed were therefore related by the design, rather than varied independently.

Table 1. Nominal stimulus settings supplied by the laboratory; every category occurred on both sides. [Jason: Are the starting-position values in mm, and relative to which reference?]

Category	Starting-position setting	Target-speed range
1	120	255-300 mm/s
2	140	298-350 mm/s
3	160	340-400 mm/s

The experiment included central-fixation and free-eye-movement conditions. This study analyzed only the free-eye-movement condition (condition 2), as specified by the laboratory. Within this subset, the modeling variables were starting-position category, starting side and executed target speed.

1.2 Recordings and analysis sample

Finger position was recorded in three dimensions at 240 Hz using a Polhemus Liberty magnetic motion-capture system. Each CSV contained positions, sensor orientation, a frame counter, a reception timestamp and an event marker. Orientation was excluded. Frame counters supplied the sampling time base because reception times need not follow a regular sampling interval. Paired MATLAB files supplied the target trajectory at 60 Hz, event times and outcome labels.

The available condition-2 dataset comprised 4,763 trials from 28 participants. Four trials without an arrival event and 27 with arrival more than one second after the target window ended were excluded, leaving 4,732 trials. Early-arrival and fixation-warning labels were retained. [Jason: Confirm the late-arrival exclusion and retention of fixation-warning trials in condition 2.]

Twenty-six participants contributed 171-180 retained trials each; the other two contributed 60 and 28. These two participants had only 60 and 29 recordings available, respectively. Thus, the small counts mainly reflect incomplete available sessions rather than extensive exclusion. The median retained count was 179.5. [Jason: Are the missing recordings for subject02 and subject41 unavailable files or incomplete sessions, and should both participants remain in the analysis?]

1.3 Task and recorded-movement examples



Figure 1. Original free-eye-movement instruction screen supplied with the stimulus files (instructions2.jpg). The lower red square indicates the starting region; the upper blue square is the interception region. The instructions require finger arrival while the moving target occupies that region and permit free eye movements. This is the task display, not a photograph of the apparatus.

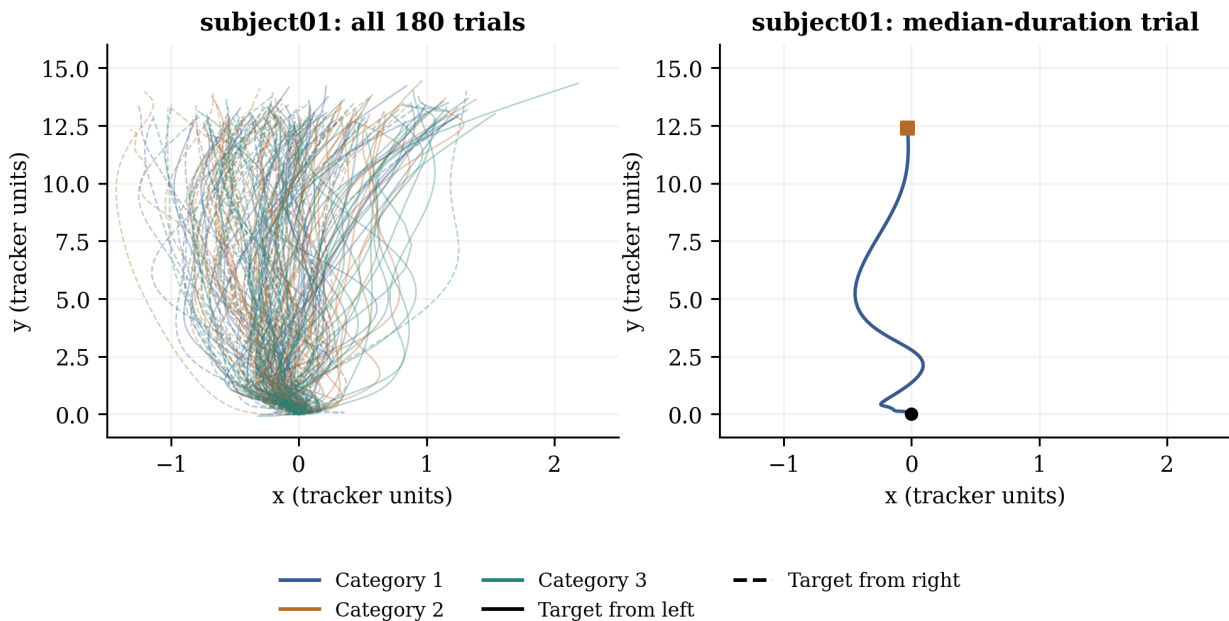


Figure 2. Recorded movements from subject01, selected as the first participant in identifier order. Left: all 180 retained trials, colored by target starting category; line style indicates target side. Right: one trial selected by median modeled duration, before inspecting model errors. Positions are filtered, aligned to target motion and translated to a common starting point. Panels share axis limits, with lateral variation enlarged relative to forward displacement. These are recorded data, not model outputs.

Repeated trials provide the observations from which a participant-specific distribution can be estimated. The spatial paths illustrate variation in the recordings, while the subsequent evaluation also examines initiation time, movement duration and derived kinematic features. Figure 3 shows how an individual recording enters the models.

2 From recordings to model inputs

Trial durations and the number of recorded samples varied. To compare and compress trajectories with a common model, we mapped every retained recording to 100 positions spanning the same relative interval within the trial. This temporal normalization was applied after establishing a regular sampling grid, filtering and selecting the modeled window.

2.1 Sampling grid, filtering and event alignment

Samples sharing a frame counter were averaged, and missing counters were filled by linear interpolation. The resulting grid had a 1/240 s interval. A fourth-order Butterworth low-pass filter with a 10 Hz cutoff was applied forward and backward to the three position coordinates. Filtering attenuated rapid fluctuations before resampling and speed-based onset detection.

Marker 5 identified target appearance. Target-motion onset was located from the first nonzero displacement in the executed target trajectory and mapped to the nearest tracker frame. The modeled window ran from that frame to the final recorded tracker sample. It therefore included waiting after target motion began. [Jason: Confirm the target-motion alignment and whether the earlier appearance-to-motion interval should also be modeled.]

The final tracker sample served as the arrival proxy; the event audit placed it, on average, 26.2 ms before the MATLAB arrival event (SD 3.6 ms). [Jason: Confirm the arrival-event meaning and whether this endpoint is appropriate.]

2.2 Temporal and spatial normalization

For a window containing L samples, sample i was assigned phase $u_i = i/(L - 1)$. A cubic interpolant S was fitted through the filtered coordinates and evaluated at 100 phases $u_j = j/99$, including both endpoints. Subtracting the first position then aligned the start of every trajectory:

$$\mathbf{P}_j = \mathbf{S}(j/99) - \mathbf{S}(0), \quad j = 0, \dots, 99.$$

This operation changes the sampling grid without fitting the later five-knot model. A phase of 0.5 denotes halfway through the selected time interval, not halfway along the traveled path. The resulting points preserve relative temporal progression, but the array alone does not specify duration in seconds. Spatial normalization translates the path; it does not rotate it or scale its amplitude.

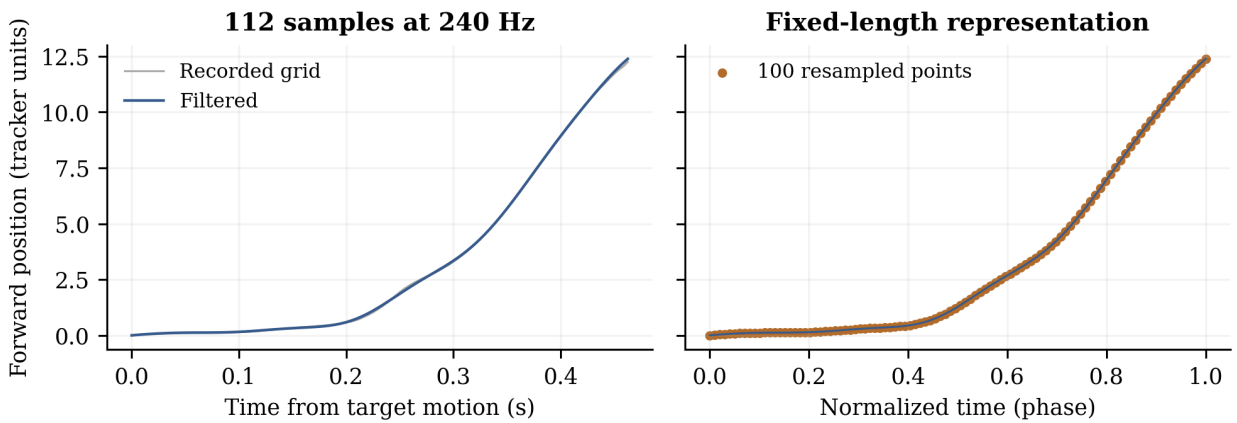


Figure 3. One subject01 trial before and after temporal normalization. The filtered forward-position trace uses physical elapsed time on the left. The right panel shows the same trace against phase, with all 100 resampled points. The same operation is applied to the other coordinates. Resampling does not create 100 independent observations.

The analysis used x and y from the normalized 3-D trajectory, producing a 100×2 matrix $\mathbf{P}$, flattened in time order into 200 neural input values. [Jason: Confirm that x - y is the task plane and confirm tracker position units (mm or cm).] Until confirmed, trajectory errors are reported in squared tracker units.

2.3 Duration targets and participant separation

Physical timing was retained separately from the phase-normalized coordinates. Finger onset was the first post-target-motion sample with 3-D speed above 5 tracker units/s for three consecutive frames. Initiation time was measured from target motion to this onset; movement time was measured from onset to the final sample. If no sustained crossing was found, the implementation used target motion as onset. [Jason: Confirm the onset threshold, persistence rule and fallback.]

The two durations were transformed as $\log(t + 0.001)$, with t in seconds, and standardized using training-trial means and standard deviations. They were supervised prediction targets and were not supplied as encoder inputs. Predictions were converted back to seconds by reversing standardization and the log transform, with negative values clipped to zero.

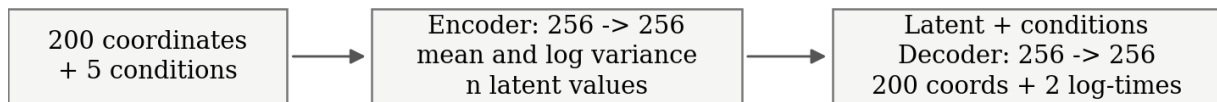
Four fixed participant folds contained 17 training, four validation and seven test participants each. Every participant appeared in a test fold once. Parameters and normalization were fitted on training participants; validation participants determined hyperparameters and checkpoints. Neural training used inverse-trial-count sampling to give each training participant equal expected sampling weight. Reported errors were averaged within each test participant before averaging participants equally.

Reconstruction used each complete held-out trajectory as an input and compared its decoded output with that trajectory. For participant fingerprints and generation, test trials were instead divided into disjoint, condition-stratified context and query sets. Only context codes contributed to the fingerprint; query movements supplied the evaluation distribution. These distinct input requirements were maintained throughout the comparisons.

3 Compact trajectory models

Spline + PCA and the neural encoders provided alternative mappings from the same observed trajectory to n latent values. The primary comparison used $n=3$, with $n=8$ testing a less restrictive representation. CVAE and spline runs also included $n=2$ and $n=4$; the matched neural ablations were evaluated at three and eight dimensions. The spline model supplied a smooth linear representation, while the neural models learned nonlinear encoder-decoder mappings.

Conditional variational autoencoder



Spline + PCA

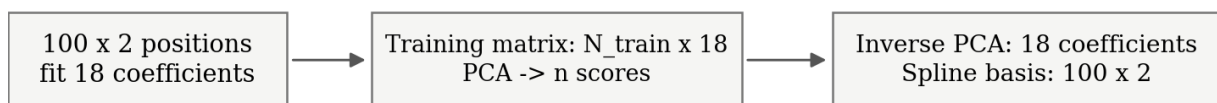


Figure 4. Alternative model pipelines. Each representation compresses a single trial to n values. A participant fingerprint is subsequently formed by averaging context-trial codes. The spline timing regression is separate from its shape reconstruction.

3.1 Spline fitting and PCA compression

The spline representation first replaced each 100-point coordinate sequence with a fitted smooth curve. We used cubic B-splines with five fixed interior knots at phases 1/6, 2/6, 3/6, 4/6 and 5/6. These define nine basis functions per coordinate. The knot locations were shared across trials; the fitted coefficients varied between trials.

Let B be the 100×9 matrix of basis values at the resampling phases. For each trial, the nine x coefficients and nine y coefficients were obtained by least-squares fitting:

$$\mathbf{c}_x = \arg \min_{\mathbf{c}} \|\mathbf{x} - B\mathbf{c}\|_2^2, \quad \mathbf{c}_y = \arg \min_{\mathbf{c}} \|\mathbf{y} - B\mathbf{c}\|_2^2.$$

Concatenating the coordinate coefficients produced an 18-value vector $\mathbf{c}$ per trial. Across training trials, these vectors formed an $N_{\text{train}} \times 18$ matrix. PCA was fitted to this matrix to learn a shared n -dimensional coefficient subspace. Raw and coefficient-standardized variants were compared using validation trajectory MSE, and the lower-error variant was retained. The PCA basis and scaling statistics were never fitted to test participants.

Writing $\mathbf{m}$ for the training coefficient mean, $\mathbf{d}$ for the coefficient scaling vector and V for the $18 \times n$ principal-component matrix, encoding and reconstruction can be expressed as:

$$\mathbf{z} = V^T[(\mathbf{c} - \mathbf{m}) \oslash \mathbf{d}], \quad \hat{\mathbf{c}} = \mathbf{m} + \mathbf{d} \odot (V\mathbf{z}).$$

Division and multiplication by $\mathbf{d}$ act coordinate-wise; $\mathbf{d}$ equals one for the unstandardized variant. PCA selects directions of variance in coefficient space. Its fitting objective is therefore distinct from minimizing trajectory MSE directly, although trajectory MSE is used to select the scaling variant.

The reconstructed coefficient vector was separated into x and y blocks. Multiplying each reconstructed coefficient block by B recovered its coordinate sequence. Test trajectories were fitted to the fixed spline basis to obtain coefficients and then projected through the training-fitted PCA basis. Thus, the final trial code contained n values, not all 18 spline coefficients.

3.2 Ridge regression for spline timing

Shape-only PCA did not contain explicit duration targets. We therefore fitted a separate Ridge regression to predict the two standardized log-durations from the concatenation of the n PCA scores and five task-condition values. Its input matrix had N_{train} rows and $n+5$ columns, standardized using training statistics. For input matrix U and target matrix Y , the fitted weights W and intercept b minimized:

$$\|Y - UW - \mathbf{1}b^T\|_F^2 + \alpha \|W\|_F^2.$$

The penalty reduced the size of regression coefficients; the intercept was unpenalized. Alpha was selected from 0.01, 0.1, 1, 10 and 100 by validation MSE in standardized log-time space. The selected regression was fitted on training participants and inverted to seconds at prediction time. This regression predicts timing; inverse PCA and the spline basis reconstruct the spatial trajectory.

3.3 Neural encoder-decoder models

The conditional variational autoencoder (CVAE) learned a nonlinear trajectory representation within the variational encoder-decoder framework [1,2]. The 200-coordinate input was standardized using a separate training mean and standard deviation for each coordinate. Five condition values were concatenated to it: three one-hot starting-category indicators, a binary starting-side value and continuous executed target speed. Speed was calculated from the target trajectory and divided by 1920 pixels to express it in screen widths/s. [Jason: Confirm the 1920-pixel screen-width assumption.]

Two fully connected layers of 256 ReLU units produced an n-dimensional posterior mean and log-variance vector. Training drew a latent sample through Gaussian reparameterization. The decoder concatenated z with the five conditions and applied two further 256-unit ReLU layers. Separate linear heads predicted 200 standardized coordinates and two standardized log-durations.

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

The per-trial objective combined coordinate reconstruction, duration prediction and regularization of the approximate posterior toward a standard-normal prior:

$$\mathcal{L} = \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + 20\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 + \beta D_{\text{KL}}[q(\mathbf{z} | \mathbf{v}, \mathbf{c}) \| \mathcal{N}(\mathbf{0}, \mathbf{I})].$$

Here $\mathbf{v}$ denotes standardized trajectory coordinates, $\mathbf{y}$ the two standardized log-duration targets, and $\mathbf{c}$ the condition vector. Coordinate and timing errors are sums over their respective outputs; the complete loss is averaged over the batch. For the diagonal Gaussian posterior, the KL term sums contributions from all n coordinates. Beta increased from zero to one in 50 increments, reaching one at epoch 51.

$$D_{\text{KL}} = \frac{1}{2} \sum_{k=1}^n (\mu_k^2 + \sigma_k^2 - 1 - \log \sigma_k^2).$$

Training used Adam with initial learning rate 0.001, batches of 64 and gradient clipping at norm 5. Validation controlled learning-rate reduction and checkpoint selection using the full-target-KL objective. Training stopped after 25 epochs without improvement or at 150 epochs. Seeds 42, 43 and 44 repeated training. For deterministic reconstruction, the decoder received the posterior mean rather than a sampled code; its outputs were then returned to the original coordinate and timing scales.

Two ablations isolated changes to this architecture. Conditional AE (CAE) retained the condition inputs but replaced stochastic sampling with a deterministic code and removed the KL term. Unconditional VAE retained sampling and KL but replaced all condition inputs with zeros at both encoder and decoder. Its mathematical formulation consequently omitted the dependence on $\mathbf{c}$. The same trajectory and timing targets were used in these comparisons.

3.4 Condition-only control and dashboard scope

A separate condition-only Ridge model predicted the 200 trajectory coordinates from the five task values. It received neither the observed trajectory nor a participant fingerprint, providing a reference for prediction from task information alone. This differs from the spline timing regression, whose inputs include a trajectory code.

The dashboard supports CVAE and spline + PCA at $n=2, 3, 4$ and 8, and VAE and CAE at $n=3$ and 8. VAE at $n=8$ is the default for distribution exploration because it has the lowest cohort-average KS, energy and MMD-squared distances in the tested matrix (Section 5). This endpoint-specific numerical ranking does not imply superiority on every task. Conditions are inactive for VAE and affect only timing for spline + PCA. No $n=16$ result is included in the matched study.

4 Trajectory reconstruction results

The first evaluation asked how much of an observed trajectory each compact representation could recover. Both the spline and neural models received the complete held-out trajectory, compressed it to n values and decoded it to the original 100×2 shape. Reconstruction MSE was calculated as:

$$\text{MSE} = \frac{1}{200} \sum_{j=0}^{99} [(x_j - \hat{x}_j)^2 + (y_j - \hat{y}_j)^2].$$

Trial errors were averaged within participants, then across neural seeds and finally across the 28 participants with equal weight. This preserves each participant as the reporting unit despite unequal trial counts. Spline fits were deterministic within each fold and dimension. Errors are in squared tracker units.

Table 2. Participant-balanced reconstruction error. Lower is better. Condition-only Ridge has no latent dimension; its result is repeated for reference.

Model	n=3 MSE	n=8 MSE
Spline + PCA	0.097255	0.022772
VAE	0.148299	0.033487
Conditional AE	0.169661	0.025968
CVAE	0.176981	0.032481
Condition-only Ridge	1.233795	1.233795

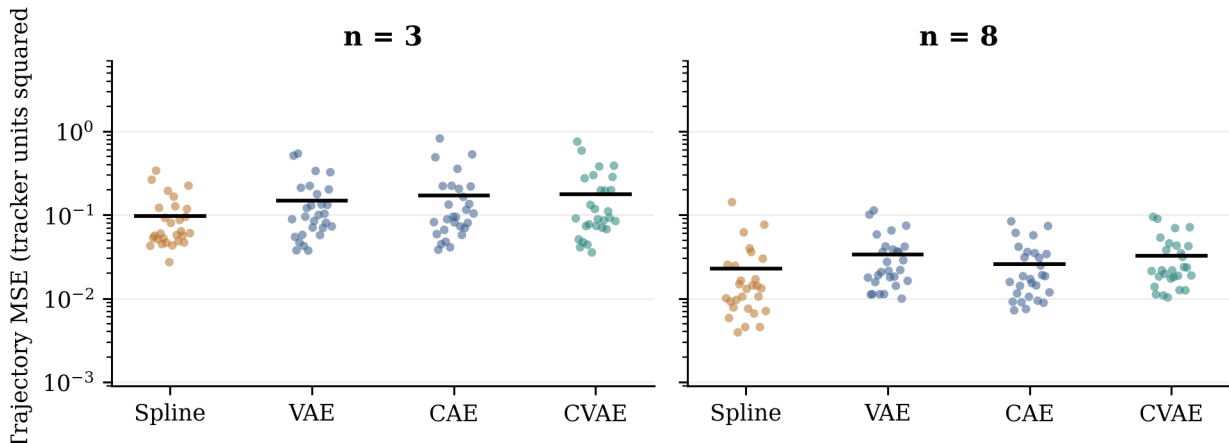


Figure 5. Participant errors after neural-seed averaging. Each point represents one participant; black bars show cohort means. The vertical axis is logarithmic.

Spline + PCA achieved the lowest mean reconstruction error at both capacities. Among neural models, VAE was best at $n=3$ (0.148299), whereas CAE was best at $n=8$ (0.025968). CVAE error decreased from 0.176981 to 0.032481, and spline error decreased from 0.097255 to 0.022772, when capacity increased from three to eight values.

Exploratory two-sided paired Wilcoxon tests used 28 seed-averaged participant values, Pratt zero handling and differences rounded to 12 decimals [3]. Benjamini-Hochberg adjustment covered 140 comparisons: ten model pairs, two dimensions and seven endpoints. At $n=3$, VAE had lower MSE than CVAE ($q=0.00163$) and CAE ($q=0.03695$). At $n=8$, CAE had lower MSE than both variational models ($q<0.000001$). All six spline-neural contrasts favored spline after this adjustment.

Holm adjustment across the same 140 tests retained the two $n=8$ neural contrasts but not the two $n=3$ contrasts. Because folds share development participants, these tests provide exploratory within-cohort evidence rather than independent population confirmation [4].

4.1 Recorded and reconstructed trajectories

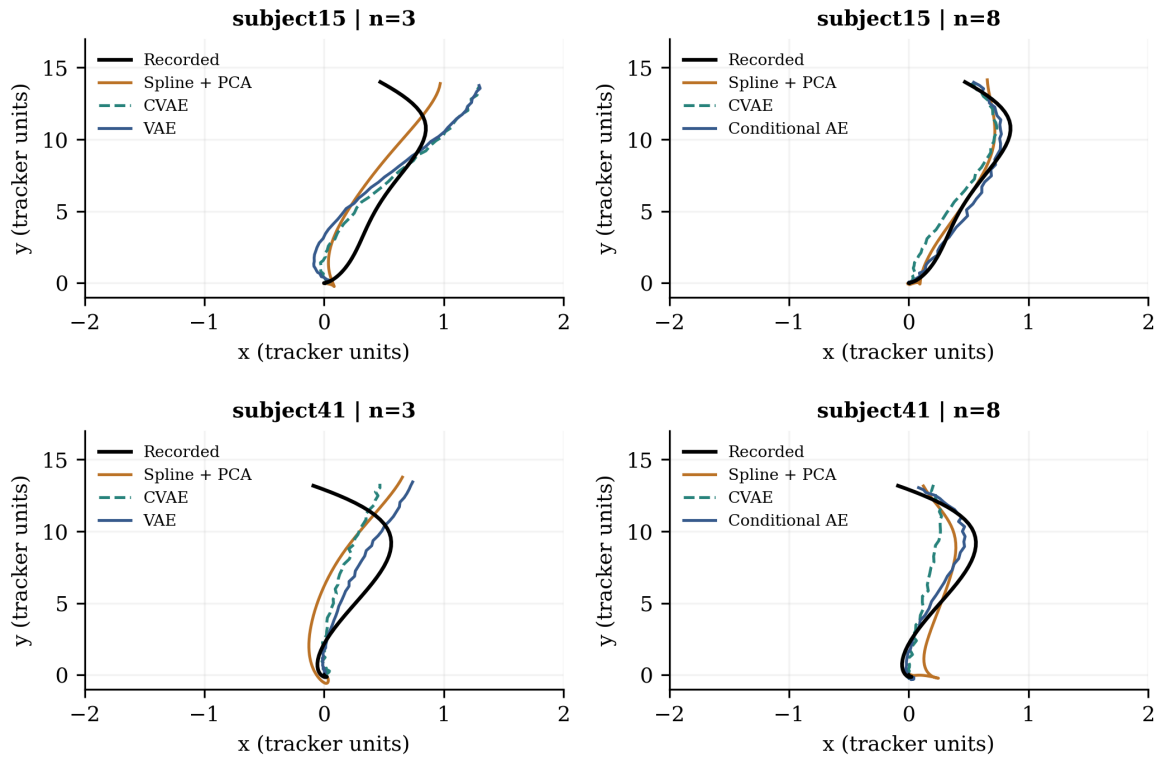


Figure 6. Recorded inputs and decoded outputs, using neural seed 42. Blue represents VAE at $n=3$ and CAE at $n=8$. All panels share axis limits; the lateral axis is expanded relative to the forward axis to reveal differences. These are reconstructions, not newly generated movements.

Examples were chosen by a fixed error-ranking rule. Participants were ranked by mean $n=3$ reconstruction MSE across the four compression families, averaging neural seeds first. The 14th and 26th participants represented the central and higher-error portions of the ranking. Within each participant, the median-ranked trial under the same family-averaged error was selected. The same input was displayed at both capacities.

The selected trials were **subject15_li_2_1_1_26** (fold 2); **subject41_li_2_1_1_25** (fold 1). Selection used held-out errors for illustration only. The aggregate comparison includes all 28 participants; a model with the lowest cohort mean need not be best on each displayed trial.

Reconstruction establishes how accurately the models compress observed paths. The fingerprint objective requires a further test: whether a code estimated from one set of a participant's movements can reproduce the distribution of other movements. The subsequent sections therefore evaluate duration prediction, generated feature distributions and participant-specific information. Side-by-side generation figures will be labeled separately from these input-output reconstructions.

5 Generating participant-specific movement distributions

Accurate reconstruction shows that a code retains information about an observed movement. The fingerprint objective asks whether a code obtained from a participant's previous trials can also describe their other movements. We therefore evaluated generation using disjoint context and query trials from participants excluded from model development. Because generated movements have no unique recorded counterpart, this evaluation compared distributions rather than assigning a pointwise reconstruction error to each generated trajectory.

5.1 Context enrollment and stochastic generation

Within each held-out participant, trials were divided approximately equally into context and query sets, stratified by starting category and side. A fixed partition was shared across models. Each context trajectory was encoded, and its latent code contributed to the participant fingerprint. For variational models the code was the posterior mean; CAE used its deterministic encoding and spline + PCA used its PCA scores. The fingerprint was the mean of these context codes:

$$\theta_s = \frac{1}{|C_s|} \sum_{i \in C_s} \mu_i, \quad \mathbf{z}^{(b)} \sim \mathcal{N}(\theta_s, \Sigma_{\text{train}}).$$

Here C denotes the participant's context set and the fingerprint has n values. Trial variability was supplied by a shared covariance estimated from training participants, allowing multiple latent samples to be drawn around that fingerprint. For the neural models, this covariance combined variation in trial codes around each training participant's mean with the mean encoder posterior variance:

$$\Sigma_{\text{train}} = \text{Cov}_{\text{train}}(\mu_i - \bar{\mu}_{s(i)}) + \text{diag}(\mathbb{E}_{\text{train}}[\sigma_i^2]) + 10^{-6}I.$$

The small diagonal term stabilizes sampling. For CAE, the posterior-variance contribution is approximately zero. Spline + PCA used the covariance of training scores after subtracting each training participant's mean, with the same diagonal stabilizer. Consequently, individual context trials determine the fingerprint center, while the covariance is shared. The n -value fingerprint does not independently estimate each participant's variability or covariance structure.

For each participant and fitted model, 120 latent samples were decoded into trajectories and durations. CVAE and CAE received task-condition vectors drawn from the participant's query-condition mixture; VAE ignored those vectors. Query task metadata supplied the mixture being evaluated, while query trajectories were withheld from fingerprint construction and generation. For spline + PCA, sampled scores were decoded through inverse PCA and the spline basis, and the timing Ridge received those scores and conditions. Its spatial decoder did not depend on conditions.

The generated outputs were therefore complete phase-normalized trajectories accompanied by predicted movement and initiation times. The evaluation reflects generation under the observed condition mixture. It does not establish extrapolation to new task conditions or adaptation across recording sessions.

5.2 Movement features

We extracted 11 timing, speed and geometry features (Table 3). Recorded features used the observed movement-onset-to-end segment. Generated segments used predicted initiation as a fraction of total duration, capped at 0.98, followed by resampling to 100 points. Movement duration was floored at 1 ms; initiation time at zero.

5.2 Movement features (continued)

Table 3. The 11 generated-feature endpoints. Speed-peak count is a heuristic descriptor and is distinct from the minimum-jerk component count evaluated later.

Feature	Definition and unit
Initiation time	Target-motion onset to finger onset; seconds.
Movement time	Finger onset to the final sample; seconds.
Peak speed	Maximum norm of the position gradient; tracker units/s.
Time to peak speed	Peak-speed sample index divided by the number of movement intervals; fraction.
Path length	Sum of consecutive position-step lengths; tracker units.
Straight-line distance	Distance from movement start to movement end; tracker units.
Curvature index	Path length divided by straight-line distance; dimensionless.
Maximum lateral deviation	Largest perpendicular distance to the start-to-end line; tracker units.
Speed-peak count	Peaks with prominence at least 10% of the maximum and separation of about 50 ms; minimum reported count is one.
Endpoint x; endpoint y	Two features: final movement coordinates relative to movement onset; tracker units.

Both recorded and generated movement segments passed through the same feature function. Positions were low-pass filtered with a cutoff of min(10 Hz, 0.45 times the effective sampling rate), and finite differences produced velocity. The effective rate was 99 divided by movement duration in seconds. The feature code was shared, although recorded onset and timing were measured and generated onset and timing were predicted. The recorded segments also inherited acquisition preprocessing; this comparison does not remove that upstream difference.

5.3 Distribution comparisons

For each participant and feature, the two-sample Kolmogorov-Smirnov statistic [3] measured the largest difference between the recorded-query and generated empirical cumulative distributions. The mean of the 11 statistics summarized marginal feature fidelity:

$$D_{sf} = \sup_x |F_{sf}(x) - G_{sf}(x)|, \quad \overline{D}_s = \frac{1}{11} \sum_{f=1}^{11} D_{sf}.$$

Lower values indicate closer distributions. We used KS as a descriptive distance, including for discrete peak counts; continuous-distribution KS p-values were not used to infer fidelity of that discrete feature. Matching marginal distributions alone does not ensure that relationships among features are reproduced.

Energy discrepancy and squared maximum mean discrepancy (MMD) [5] therefore compared complete 11-feature vectors. Features were centered and scaled using training data, with a scale floor of one for peak counts. The energy statistic was twice the average cross-sample Euclidean distance minus the two average within-sample distances:

$$E = 2 \overline{\|X - Y\|} - \overline{\|X - X'\|} - \overline{\|Y - Y'\|}.$$

These averages included all ordered pairs and within-sample diagonal zeros; the reported statistic was not square-rooted. MMD used a Gaussian kernel with bandwidth fixed from training data and excluded diagonal within-sample pairs:

$$\widehat{\text{MMD}}_u^2 = \frac{\sum_{i \neq j} k(X_i, X_j)}{m(m-1)} + \frac{\sum_{i \neq j} k(Y_i, Y_j)}{r(r-1)} - \frac{2 \sum_{i,j} k(X_i, Y_j)}{mr}.$$

Here m and r are recorded and generated sample counts. The kernel was exp(-gamma times squared Euclidean distance), with gamma set to the inverse median positive squared distance among a fixed training subset of up to 512 trials. This unbiased MMD-squared estimator can be slightly negative in finite samples. All distance references were shared across models within a fold.

5.4 Generated-feature results

Distances were computed separately for each test participant, averaged across available neural training seeds and then across the 28 participants with equal weight. Table 4 presents the four compact-representation families at matched capacities. Spline + PCA contributes a single deterministic fit and fixed sampling realization per fold and dimension; neural results summarize seeds 42, 43 and 44.

Table 4. Cohort-average generated-feature discrepancies. Lower is better. All three metrics compare distributions; none is a trajectory reconstruction MSE.

Model	n	Mean KS	Energy	MMD squared
Spline + PCA	3	0.325126	0.977433	0.181467
CVAE	3	0.277795	0.745431	0.130935
CAE	3	0.287578	0.801489	0.141410
VAE	3	0.284587	0.747566	0.130298
Spline + PCA	8	0.254794	0.734715	0.115291
CVAE	8	0.219404	0.430451	0.067493
CAE	8	0.258686	0.620224	0.111135
VAE	8	0.215121	0.411014	0.063881

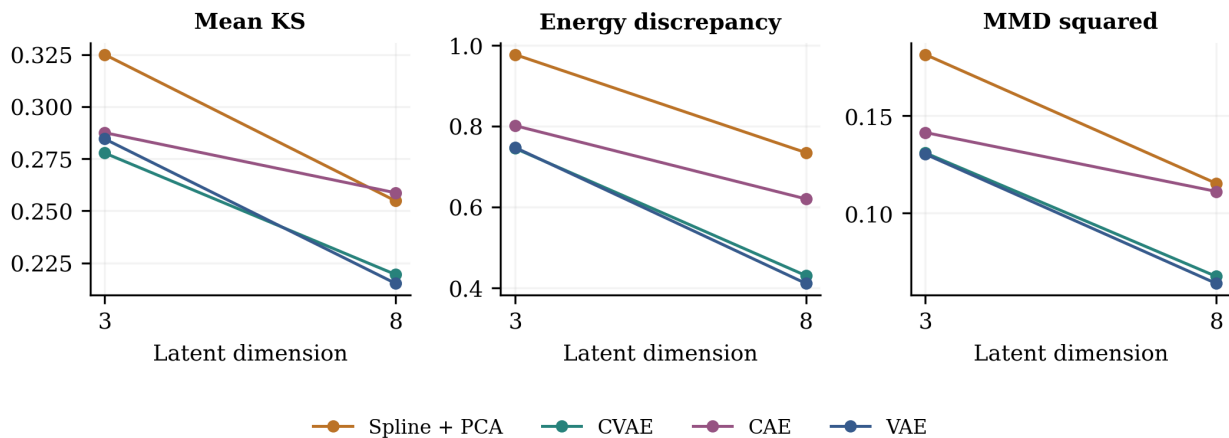


Figure 7. Generated-feature discrepancies at three and eight dimensions. Lines connect tested settings only and do not estimate performance at untested capacities.

The reconstruction ranking did not carry over unchanged to generation. At $n=3$, CVAE had the lowest mean KS (0.277795) and energy discrepancy (0.745431) among these four families, while VAE had the lowest MMD-squared value (0.130298). At $n=8$, VAE had the lowest mean on all three distances: KS 0.215121, energy 0.411014 and MMD squared 0.063881. Spline + PCA retained its reconstruction advantage but had larger generation discrepancies than VAE and CVAE at $n=8$.

Numerical advantages were not uniformly significant. Within the same exploratory 140-comparison family used for reconstruction, the $n=8$ VAE-CVAE contrast survived Benjamini-Hochberg adjustment for energy ($q=0.02575$), but not for mean KS ($q=0.13686$) or MMD squared ($q=0.05182$). None of these three contrasts survived Holm adjustment across 140 tests. The results support an endpoint-specific numerical preference rather than universal dominance.

VAE $n=8$ was consequently selected as the dashboard's default for marginal-distribution exploration. All other supported references remain selectable. The choice uses this evaluated cohort and has no separate post-selection validation. Three-dimensional models remain available for the original compact-fingerprint question; $n=16$ was not evaluated in the matched study. A higher-capacity model might improve a particular endpoint, but the present data do not establish that outcome.

5.5 Recorded and generated movement examples

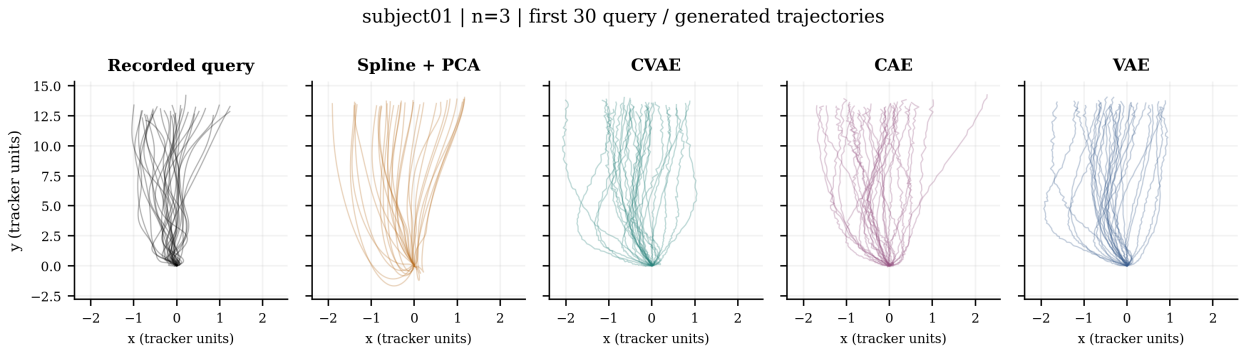


Figure 8. Recorded query trajectories and generated trajectories at $n=3$ for subject01. Each panel displays the first 30 available samples in its fixed ordering. Generated panels use context fingerprints, training-derived covariance and the seed-42 models. The recorded paths were not reconstruction inputs.

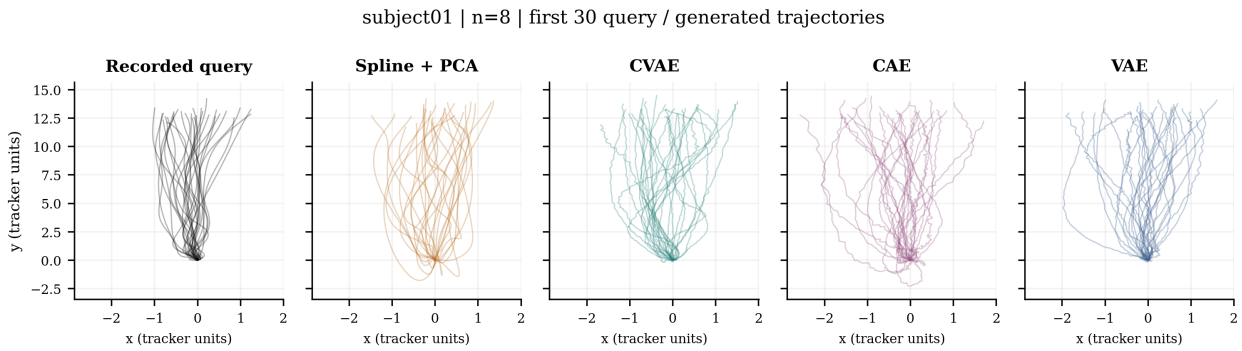


Figure 9. The corresponding $n=8$ samples for the same participant. Axis limits are shared across models within each figure and include every displayed point; Figures 8 and 9 can have different limits. Axes are not drawn at equal physical scale. No generated sample was selected for resembling a recorded trial.

The participant was selected as the first identifier in the analyzed cohort, independently of reconstruction or distribution scores. This participant was held out in fold 2. Each model generated 120 trajectories for evaluation; 30 are displayed to limit overplotting. The panels illustrate the range of decoded paths, while Table 4 summarizes the entire cohort and all evaluated seeds. Visual resemblance of individual examples is not evidence that two stochastic samples should coincide point by point.

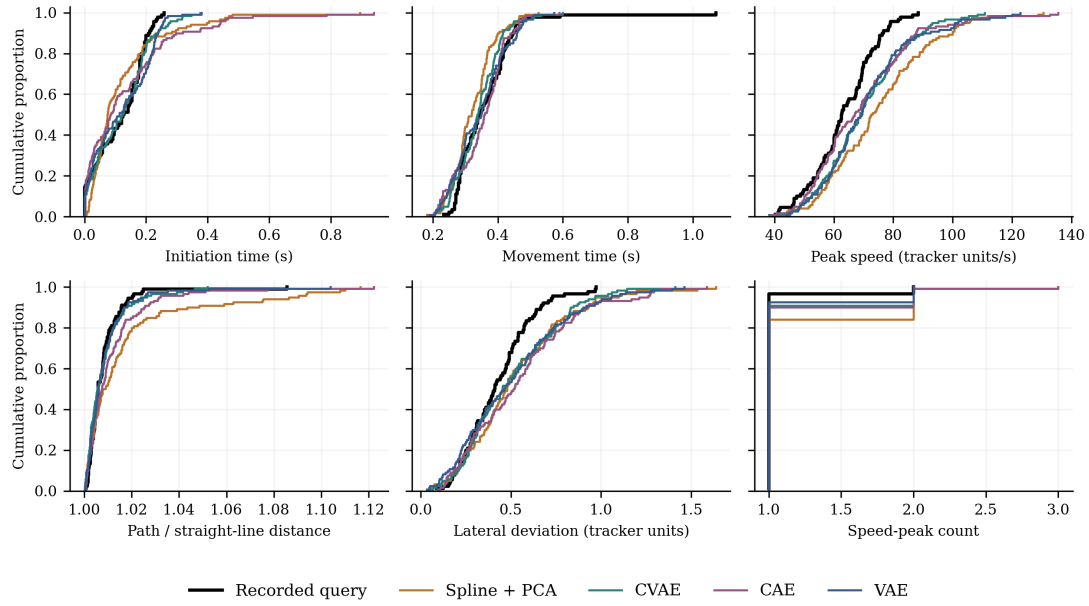


Figure 10. Six feature distributions for subject01 at n=8: recorded query trials and all 120 generated samples per model, using seed 42. These fixed display features cover timing, speed, geometry and peak count. Vertical separation between two cumulative curves illustrates a distribution discrepancy; Table 4 averages all 11 features and all participants.

These comparisons assess generation from personal context, but they do not yet isolate the contribution of the personal fingerprint. The next analysis holds the decoder and sampling procedure fixed while replacing the participant's context fingerprint with population and wrong-participant controls.

6 Testing the contribution of the personal fingerprint

The preceding experiment used a participant's context trials to generate a distribution for that participant. To establish whether the personal code contributed useful information beyond the shared decoder, we repeated generation after replacing only the center of the latent sampling distribution. This control separates the value of personal enrollment from the model's general ability to reproduce typical movements.

6.1 Fingerprint replacement within the same decoder

Three alternatives were evaluated: the participant's own context centroid; the equal-person mean of the 17 training participants' context centroids; and each of the six other test participants' context centroids. For a given participant and checkpoint, the decoder, shared covariance, 120 latent-noise draws, task-condition draws and training feature-distance reference were held fixed. The other-person result was the mean of six separately calculated distances. Their samples were not pooled into one mixture.

The control was run for all CVAE checkpoints at $n=3$ and $n=8$ (four folds and three seeds), and subsequently for all 12 VAE $n=8$ checkpoints. Each own-centroid arm reproduced its original generation distances before the substitutions were compared. Distances were averaged over training seeds within each participant and then across the 28 participants.

Table 5. Fingerprint replacement with the generative procedure held fixed. Lower discrepancies are better. The other-person row averages donor-specific distances.

Model	Fingerprint center	Mean KS	Energy	MMD squared
CVAE $n=3$	Own participant	0.278	0.745	0.131
CVAE $n=3$	Training-population mean	0.311	1.016	0.175
CVAE $n=3$	Other test participants	0.338	1.273	0.220
CVAE $n=8$	Own participant	0.219	0.430	0.067
CVAE $n=8$	Training-population mean	0.290	0.919	0.149
CVAE $n=8$	Other test participants	0.317	1.196	0.199
VAE $n=8$	Own participant	0.215	0.411	0.064
VAE $n=8$	Training-population mean	0.284	0.879	0.142
VAE $n=8$	Other test participants	0.314	1.168	0.195

6.2 Results and statistical scope

Own-context generation had lower mean discrepancies than both controls in every model/capacity comparison in Table 5. The 12 CVAE contrasts used two-sided paired Pratt Wilcoxon tests and one Benjamini-Hochberg adjustment family; the largest adjusted p-value was 0.000381. The six VAE $n=8$ contrasts formed a separate follow-up family; all adjusted p-values were at most $1.5e-06$. Both analyses used seed-averaged participant values and differences rounded to 12 decimals before ranking.

At CVAE $n=3$, the own fingerprint improved mean KS for 22 of 28 participants against the population center and 27 of 28 against other-person centers. At CVAE $n=8$ and VAE $n=8$, the corresponding counts were 27 of 28 for each control. These results support useful personal information within the recorded session. They do not imply that every participant improves or that every generated feature distribution is accurate.

6.3 Participant-level effects

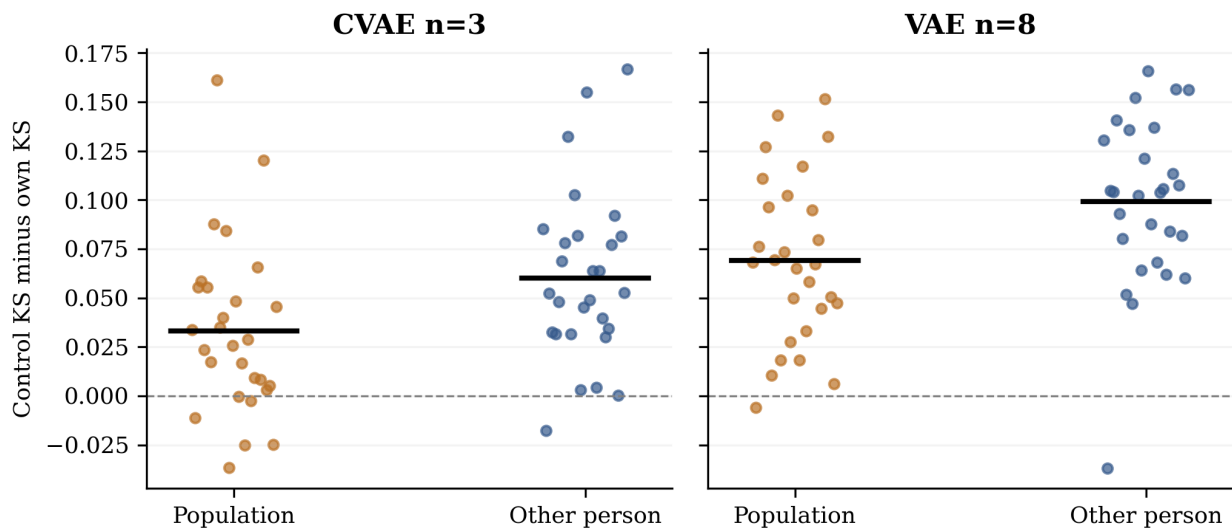


Figure 11. Change in mean KS produced by replacing a control center with the participant's own center. Positive values favor the own fingerprint. Each point is one participant after seed averaging; black bars mark means. CVAE n=3 represents the compact setting and VAE n=8 the dashboard default.

The pattern is compatible with a compact representation retaining useful person-related variation. It does not by itself establish a persistent trait: enrollment and evaluation draw from different trials within the same available session. The comparison also remains dependent on the shared decoder and covariance. There was no separate session in which a previously estimated fingerprint was tested unchanged.

These were follow-up controls on the evaluated cohort. Fold training sets overlap, and the VAE follow-up was motivated by its observed generation performance. The adjusted p-values describe these specified comparisons; they do not provide an independent validation of model selection or a physiological interpretation of the coordinates.

[Jason: Is within-session distribution modeling the intended first target, and are repeat sessions available for testing whether a participant fingerprint remains useful across sessions?]

7 Recovering physical timing from normalized trajectory shape

Resampling removes absolute duration from the trajectory input, although relative waiting and path progression remain visible. Timing prediction therefore tested whether the normalized path retained information about when movement began and how long it lasted. In this experiment the model received the complete observed test trajectory, rather than only a participant centroid. It was not forecasting the duration of a future trial before its trajectory was observed.

The neural decoders predicted two standardized log-durations alongside the coordinates. Spline + PCA used its separate Ridge timing model. Predictions were inverted to seconds and compared with durations derived from the recorded indices. Absolute errors were averaged within participants, then over neural seeds and participants; Table 6 reports milliseconds.

$$\text{MAE} = \frac{1}{S} \sum_{s=1}^S \frac{1}{N_s} \sum_{i=1}^{N_s} |t_{si} - \hat{t}_{si}|.$$

Table 6. Original timing-head errors. Physical durations were withheld from trajectory encoders. Lower participant-balanced MAE is better.

Model	n	Initiation MAE (ms)	Movement MAE (ms)
Spline + PCA	3	56.16	51.16
CVAE	3	37.83	43.45
CAE	3	37.75	44.09
VAE	3	37.92	42.91
Spline + PCA	8	55.58	50.21
CVAE	8	34.84	44.16
CAE	8	39.87	42.67
VAE	8	35.98	42.21
Condition-only Ridge	-	98.49	72.72

Spline + PCA's reconstruction advantage did not extend to its original timing head. All three neural families had smaller mean timing errors at both matched capacities. Among neural models, the lowest mean initiation error was CAE at n=3 and CVAE at n=8; VAE had the lowest mean movement-duration error at both capacities. These numerical rankings do not constitute a single overall model ranking.

7.1 Tail errors and the limits of an average

Mean absolute error does not reveal every extreme prediction. The saved CVAE n=8, seed-43 initiation predictions included an absolute error of 11.546 s, despite a cohort-level MAE on the order of tens of milliseconds. The inverse logarithmic transform is unbounded above. A one-second flag was used as a diagnostic threshold, not as a validated physiological cutoff or a rule for removing observations. Large errors remained in the reported scores.

[Jason: What timing ranges are plausible for this task, and should a future model constrain durations explicitly rather than only flagging unusually large predictions?]

7.2 Timing-head sensitivity comparisons

The original heads differed: the CVAE used a nonlinear neural timing decoder, whereas spline used linear Ridge. To examine this difference, the same two-hidden-layer MLP (64 ReLU units per layer) was fitted to each frozen representation plus its five task values. Training-only standardization was applied to codes/conditions and log-duration targets. The head used Adam at 0.001, batch size 64, L2 coefficient 0.0001 and inverse-trial-count participant weights.

Validation selected the checkpoint using weighted standardized log-duration MSE, with an improvement tolerance of $1e-7$, patience 40 and at most 400 passes. Four folds, two capacities, two representations and three seeds produced 48 small timing-head fits. Neither neural representation was retrained. In a separate sensitivity, each original timing output was multiplied by a validation-fitted factor minimizing participant-weighted absolute error in seconds. This calibration was then applied unchanged to test predictions.

Table 7. Timing sensitivities in ms. BH adjustment covers the eight paired participant comparisons shown. These saved sensitivity tests use Pratt handling without the explicit 12-decimal rounding used in the main comparison.

Head	n	Duration	CVAE MAE	Spline MAE	BH-adjusted p
Common MLP	3	initiation	41.43	45.75	0.007645
Common MLP	3	movement	43.89	46.12	0.1872
Common MLP	8	initiation	42.72	47.87	0.04228
Common MLP	8	movement	44.84	44.81	0.8314
Calibrated original	3	initiation	37.24	44.40	9.769e-05
Calibrated original	3	movement	44.33	50.93	6.056e-05
Calibrated original	8	initiation	34.49	53.56	1.192e-07
Calibrated original	8	movement	44.71	50.45	0.0001441

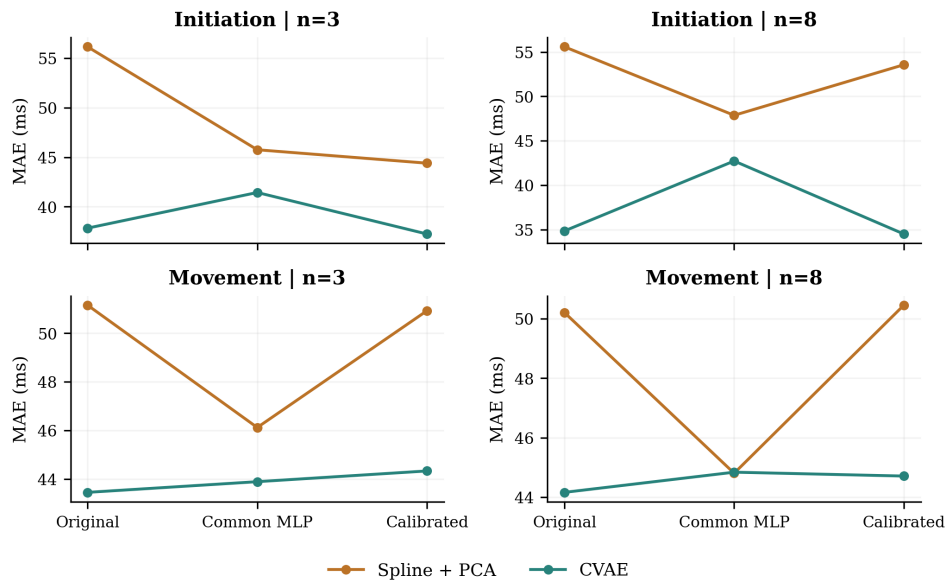


Figure 12. Timing errors with original, common-MLP and validation-calibrated heads. Points represent separate fitted procedures; lines connect them to make the comparison readable.

With the common MLP, CVAE retained lower initiation MAE after the eight-test correction, while movement-time differences did not survive correction. Calibration retained CVAE's advantage over spline on both durations. These checks do not isolate a pure representation effect: CVAE codes had been learned with timing supervision, while spline PCA had been fitted to shape coefficients. They also do not show that calibration is uniformly beneficial or establish the cause of the original gap.

8 Participant separation and behavioural information in the code

The replacement control establishes that the personal centroid matters to generation. Two further analyses asked what information could be read from the latent representation: whether query trials could be assigned to the correct enrolled participant, and whether a participant's code predicted specific behavioural summaries.

8.1 Closed-set participant enrollment

Within each outer test fold, all seven test participants were enrolled using their context centroids. Each query trial was assigned to the nearest centroid by Euclidean distance after dividing each coordinate by its standard deviation over the pooled test-context codes (plus 1e-8). Query codes did not determine this scale. Balanced accuracy averaged recall across the seven participants, and results were averaged across folds and training seeds. The reference for uniform guessing was 1/7, or 14.3%.

Table 8. Identification among seven enrolled test participants. These are descriptive closed-set results, not identification of previously unseen identities without enrollment.

Model	n=3 balanced accuracy	n=8 balanced accuracy
Spline + PCA	36.4%	48.6%
CVAE	40.9%	50.7%
CAE	39.0%	46.5%
VAE	40.0%	50.8%

Accuracy exceeded the 14.3% reference numerically for every listed setting. At n=8 the two variational models were near 51%, while at n=3 they were near 40%. This result indicates partial separation of participant-related variation, with considerable overlap. It is not evidence for perfectly separated clusters or stability across sessions.

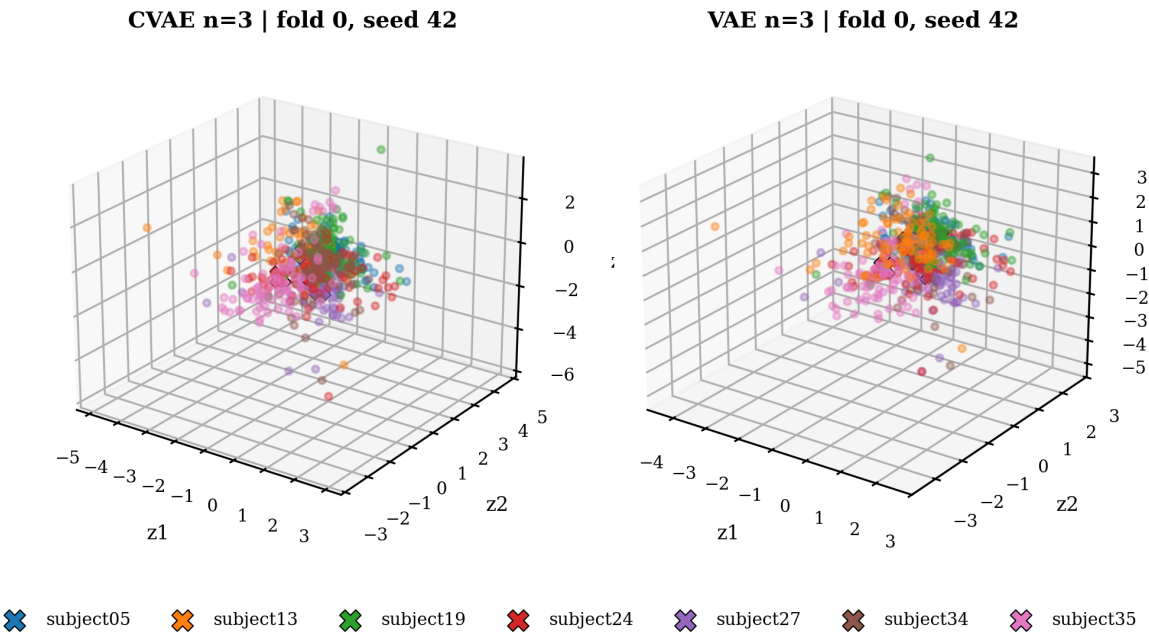


Figure 13. Three-dimensional query codes (small points) and context centroids (crosses) for all seven test participants in fold 0, seed 42. Each panel uses one fitted coordinate system. Axes from separately fitted models are not aligned; the viewing angle and point selection do not optimize apparent separation.

8.2 Earlier clustering diagnostic

An earlier descriptive analysis standardized the complete trajectory matrix or a selected kinematic-feature matrix over all available trials, then applied K-means with 28 clusters, ten initializations and seed 42. Agreement with participant labels was measured by adjusted Rand index (ARI). Two hundred permutations of participant labels provided a reference for each fixed clustering; the p-value used a plus-one correction. This analysis was in-sample and did not use the outer model-evaluation folds.

Table 9. Historical K-means diagnostic. Small ARI values indicate weak agreement with participant identity even though both exceed their permutation reference.

Input	ARI	Normalized mutual information	Permutation p
trajectory	0.0518	0.1944	0.00498
kinematic features	0.0703	0.2197	0.00498

K-means assigns observations to the nearest of 28 fitted centers; it was not given participant labels while fitting those centers. The labels were used afterward to evaluate agreement. ARI compares pairwise co-membership with chance adjustment: a value near zero indicates little agreement beyond its chance reference. Normalized mutual information measures dependence between the cluster and participant labels, but is not chance-adjusted in the same way.

The historical kinematic matrix used the configured clustering feature list, including endpoint z and excluding speed-peak count. It should not be confused with the later 11-feature generation endpoint. Standardization and clustering used all available trials for this descriptive analysis, whereas reconstruction and generation were assessed on participants held out from model fitting.

Both observed ARIs exceeded every one of the 200 shuffled-label values, giving the minimum attainable plus-one p-value of 1/201. A small permutation p-value alongside a small ARI indicates detectable but weak agreement, not clean participant clusters. Trial-label exchangeability is an assumption of this descriptive reference; repeated trials and session structure limit a broader inferential interpretation.

8.3 Predicting participant-level behavioural summaries

For each participant, the means and sample standard deviations of seven query features formed 14 targets: initiation time, movement time, curvature index, peak speed, path length, maximum lateral deviation and heuristic speed-peak count. A separate Ridge regression predicted each target from the context-mean latent fingerprint. Thus the predictor matrix contained one row per participant, rather than one row per trajectory.

Coordinate scaling was fitted on the 17 training participants. Ridge alpha was chosen from 0.01, 0.1, 1, 10 and 100 using MSE on four validation participants. After selection, the regressor was refitted on all 21 development participants while retaining the training-fitted input scale, and tested on seven participants. The encoder remained frozen. Predictions from all four test folds were pooled to score 28 participants per training seed.

$$R^2 = 1 - \frac{\sum_s (y_s - \hat{y}_s)^2}{\sum_s (y_s - \bar{y})^2}.$$

Pooled R-squared used the mean of the 28 observed query-summary targets in its denominator. Zero is the error of that constant mean; negative values indicate greater squared error. This comparison constant is distinct from a deployable training-mean predictor. Averaging seven-person fold R-squared values gives a different result because each fold has its own denominator. Figure 14 reports pooled scores averaged over available model seeds and retains all negative values.

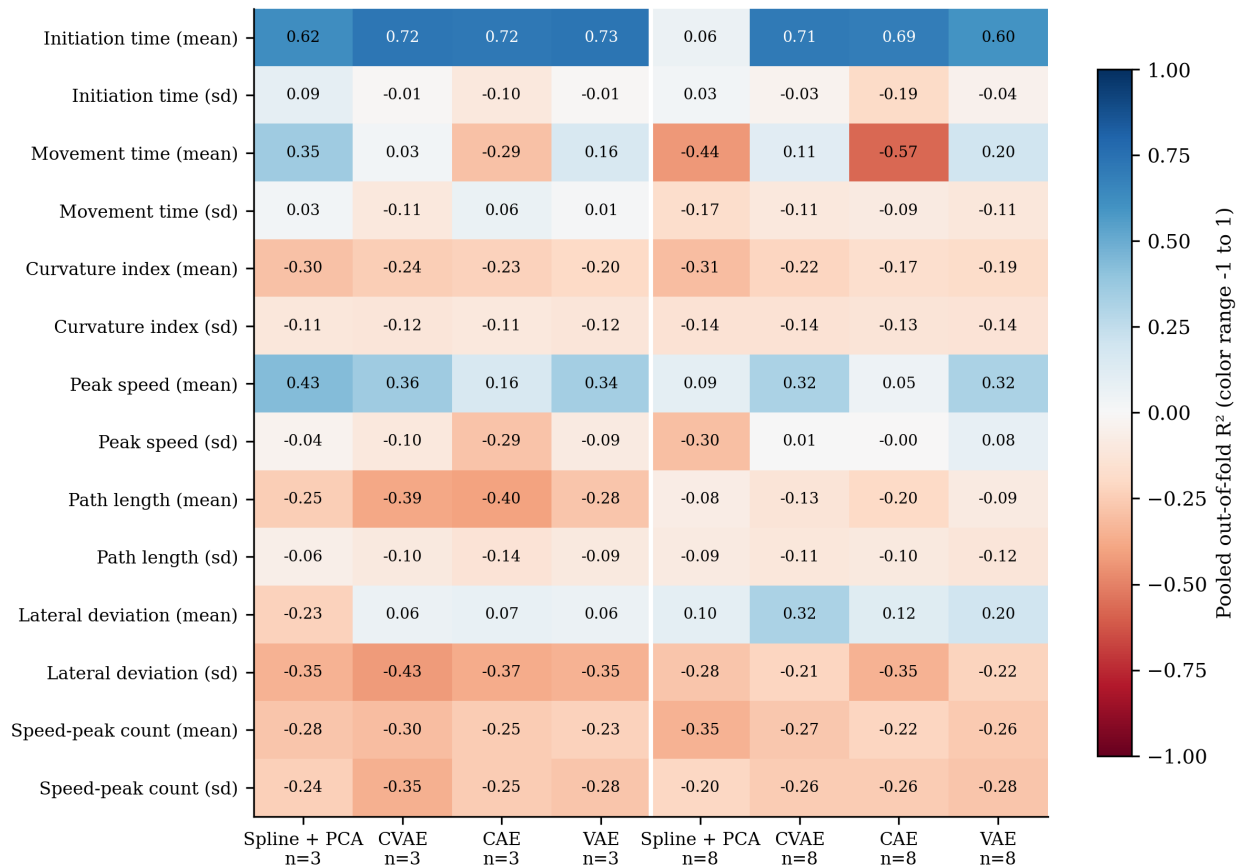


Figure 14. All 14 behavioural-summary targets for the four matched model families. Colors saturate below -1, but annotations show the actual values. No targets are removed because their R-squared is negative. Spline has one deterministic fit per fold; neural values average three training seeds.

CVAE n=3 had 10 of 14 pooled scores below zero and 12 of 14 mean-fold scores below zero. The useful personal information demonstrated by generation therefore did not translate into uniformly accurate prediction of these individual summaries.

VAE n=8 had 9 of 14 pooled scores below zero and 11 of 14 mean-fold scores below zero. The useful personal information demonstrated by generation therefore did not translate into uniformly accurate prediction of these individual summaries.

8.4 Direct context measurements as a reference

The same context recordings also allow direct measurement of each target summary. We therefore compared the latent probes with copying the corresponding context mean or sample SD to predict the query summary. A further constant predictor used the equal-person average of the 21 development participants' query summaries. These references received no test-query values. The comparison was originally tabulated for CVAE; the current report extends the same calculation to the saved CAE, VAE and spline predictions after verifying identical query targets.

Table 10. MAE for all 14 query-summary targets, highlighting the compact CVAE and dashboard VAE. Timing is in ms; other errors use each feature's units. Scores average seed-specific absolute errors, not errors after averaging predictions.

Query-summary target	CVAE n=3	VAE n=8	Direct context	Dev. constant
Initiation time (mean) [ms]	14.778	17.173	9.275	30.381
Initiation time (sd) [ms]	13.717	14.602	10.928	14.028
Movement time (mean) [ms]	36.444	35.394	11.049	48.762
Movement time (sd) [ms]	30.951	32.945	27.879	32.441
Curvature index (mean)	0.016	0.015	0.008	0.016
Curvature index (sd)	0.054	0.055	0.039	0.054
Peak speed (mean)	7.744	8.691	1.734	11.876
Peak speed (sd)	2.627	2.354	1.707	2.646
Path length (mean)	0.386	0.324	0.162	0.369
Path length (sd)	0.677	0.697	0.438	0.673
Lateral deviation (mean)	0.188	0.171	0.038	0.214
Lateral deviation (sd)	0.112	0.100	0.067	0.104
Speed-peak count (mean)	0.068	0.067	0.022	0.066
Speed-peak count (sd)	0.139	0.136	0.079	0.134

Direct context measurements yielded mean-initiation R-squared of 0.912 and mean-peak-speed R-squared of 0.976, with MAEs of 9.275 ms and 1.734 tracker units/s. These descriptive references are especially relevant when the intended output is a specific summary rather than a sampled trajectory.

Table 11. Coverage of the direct-context advantage across all saved mean-fingerprint probes. This is a descriptive comparison without a new significance test.

Latent probe	n=3: targets with lower direct MAE	n=8: targets with lower direct MAE
Spline + PCA	14/14	14/14
CVAE	14/14	14/14
CAE	14/14	14/14
VAE	14/14	14/14

Direct measurement preserves the particular statistic being predicted, whereas a latent code shares three or eight coordinates across all tasks. Fourteen direct summaries jointly require 14 numbers. The comparison therefore exposes a compression tradeoff; it does not test whether direct summaries can decode complete trajectories.

8.5 Adding latent variability to the fingerprint

An additional probe variant was saved for neural capacities $n=2$ and $n=3$. It concatenated the context mean with coordinate-wise standard deviations of the equally weighted posterior mixture. Those variances included variation between context posterior means and average posterior variance. The $n=3$ variant therefore used six predictor values and must not be presented as a three-number fingerprint.

Table 12. Pooled R-squared for all saved mean-plus-SD probe variants. CVAE2 uses two means or four mean/SD values; the other expanded variants use six values. The corresponding $n=3$ mean-only scores are in Figure 14.

Query-summary target	CVAE2 mean	CVAE2 mean + SD	CVAE3 mean + SD	CAE3 mean + SD	VAE3 mean + SD
Initiation time (mean)	0.630	0.556	0.628	0.639	0.642
Initiation time (sd)	-0.064	-0.107	-0.003	-0.055	-0.040
Movement time (mean)	-0.084	-0.149	0.201	-0.295	0.073
Movement time (sd)	-0.084	0.022	-0.362	-0.879	-0.137
Curvature index (mean)	-0.297	-0.053	-0.125	-0.098	-0.250
Curvature index (sd)	-0.151	-0.040	-0.085	-0.515	-0.175
Peak speed (mean)	0.308	0.279	0.495	0.145	0.409
Peak speed (sd)	-0.175	-0.094	0.078	-0.087	0.085
Path length (mean)	-0.331	-0.095	-0.211	-0.269	-0.252
Path length (sd)	-0.079	0.135	0.060	-0.019	0.039
Lateral deviation (mean)	-0.115	0.023	0.158	0.045	0.017
Lateral deviation (sd)	-0.403	-0.227	-0.351	-0.332	-0.377
Speed-peak count (mean)	-0.289	-0.383	-0.607	-0.562	-0.306
Speed-peak count (sd)	-0.270	-0.271	-0.805	-0.692	-0.598

At $n=3$, the number of targets with positive pooled R-squared changed from four to six for CVAE, four to three for CAE and five to six for VAE. These counts are descriptive and are not a combined performance score. Adding coordinate variability did not yield uniformly accurate behavioural prediction.

8.6 Decoder associations in the dashboard

The dashboard provides a complementary descriptive view: it samples 250 to 1,000 latent points around the training-population center with the shared covariance (sampling seed 1147), decodes them under the selected condition where supported, and computes Spearman correlations between each latent coordinate and six generated features. The features are initiation time, movement time, peak speed, path length, curvature and lateral deviation. These correlations describe co-variation under the chosen sampling distribution; they are not participant-held-out probe scores.

Latent-feature associations and slider traversals describe the fitted coordinate system. They do not identify independently validated strategy axes, and axes may rotate, change sign or exchange roles in another fit. The probe results above provide a separate test of predictive information instead of relying on a visually suggestive association.

9 Checking explicit task conditioning

The unconditional VAE's competitive performance raised two different questions: whether condition inputs reached the network correctly, and whether conditioning improved prediction of task-related differences. These questions require different checks. A working condition input does not guarantee lower error after finite training and validation selection.

9.1 Implementation and frozen-network perturbations

The audit checked all 96 saved neural checkpoints. CVAE and CAE concatenated the five-value condition vector to both encoder and decoder, and every tested condition coordinate affected their learned mappings. All 24 unconditional networks replaced these inputs by zeros and were exactly invariant to condition swaps. Enabling conditioning in a copy of each unconditional checkpoint while zeroing its condition-weight blocks preserved its outputs exactly. The conditional architecture can therefore represent the unconditional solution, although training need not find that solution.

A further perturbation shuffled complete valid condition tuples within each test participant using five fixed permutations. One arm retained the original latent code and changed only decoder conditions; another changed conditions at both encoder and decoder. The networks were not retrained. Table 13 reports participant-averaged trajectory reconstruction MSE under these perturbations.

Table 13. Frozen condition-shuffle diagnostic. All reported values come from the perturbation export; the last column counts participants with lower MSE under correct decoder conditions.

Model	n	Correct conditions	Decoder shuffled	Both shuffled	Correct better: decoder
CVAE	3	0.176981	0.180578	0.178176	25/28
CVAE	8	0.032481	0.035105	0.032506	28/28
CAE	3	0.169661	0.171582	0.170162	26/28
CAE	8	0.025968	0.026693	0.025942	27/28

Decoder-only shuffling increased mean error in all four settings. Changing both encoder and decoder produced smaller, mixed effects, including a small mean improvement for CAE n=8. Decoder-only shuffling deliberately disrupts the usual relation between latent code and condition. These findings establish sensitivity of the mappings, not a causal effect of task conditions or a controlled-generation accuracy guarantee.

9.2 Fidelity within task strata

A separate CVAE-VAE comparison generated 120 samples within each available starting-category by side stratum. The participant fingerprint and latent-noise draws were reused across strata; continuous executed speed was sampled from that stratum's query metadata. Empirical and generated mean trajectories were compared, and each feature was evaluated by KS and standardized absolute error of the median. Strata and neural seeds were averaged within participants before testing.

Table 14. Participant medians of condition-stratified mean-trajectory discrepancy. This is a comparison of mean generated and recorded paths, not individual-trial reconstruction. Holm adjustment covers the two capacities.

n	CVAE median mean-path MSE	VAE median mean-path MSE	Holm-adjusted p
3	0.1221	0.1134	0.5519
8	0.0869	0.0790	0.1325

Neither trajectory contrast survived the two-test adjustment, and none of 44 feature comparisons survived BH correction (two capacities, 11 features and two metrics). Category-3 minus category-1 and right-minus-left median contrasts were also summarized descriptively. Consequently, the available checks do not establish that CVAE controls task-specific distributions more accurately than VAE. They also do not establish that conditions are irrelevant to movement.

[Jason: Which task-specific change should controlled generation reproduce most clearly, and should that endpoint be prioritized in a future validation?]

10 Minimum-jerk components as a secondary kinematic description

The 11-feature evaluation includes a heuristic speed-peak count, but a local peak does not necessarily correspond to a separate movement component. To examine finer temporal structure, a secondary analysis approximated planar velocity as a sum of smooth minimum-jerk pulses. The purpose was to compare a specified kinematic description between recorded and generated movements, rather than infer decisions directly from a fitted count.

10.1 Pulse model and numerical fitting

Jerk is the rate of change of acceleration. The classical minimum-jerk formulation minimizes its squared magnitude integrated over a movement [6]. Our component basis used the resulting bell-shaped velocity profile, with onset t , duration T and planar displacement d for each component [7]. A component was zero outside its active interval; the fitted velocity was the vector sum of its components.

$$\mathbf{v}_k(t) = \frac{d_k}{T_k} (30u_k^2 - 60u_k^3 + 30u_k^4), \quad u_k = (t - t_k)/T_k \in (0, 1).$$

Candidates contained one to four components. Each was fitted by bounded nonlinear least squares using the trust-region reflective algorithm, Jacobian scaling and deterministic trial-specific restart seeds. The residual combined lateral velocity, forward velocity and speed, normalized by the recorded velocity energy. Speed in the prediction was the norm of the summed velocity, not the sum of component speeds.

$$E_K = \frac{\sum_i [\|\hat{\mathbf{v}}_K - \mathbf{v}\|^2 + (\|\hat{\mathbf{v}}_K\| - \|\mathbf{v}\|)^2]}{\sum_i [\|\mathbf{v}\|^2 + \|\hat{\mathbf{v}}\|^2]}, \quad \hat{\mathbf{v}}_K = \sum_{k=1}^K \mathbf{v}_k.$$

In this expression, the two equal terms in the denominator arise from the planar squared norm and the squared scalar speed; the denominator was floored at 1e-12. Velocity was padded with zeros beyond the observed movement so that components extending beyond the endpoint incurred error. This is an adapted implementation of Friedman's repository [7], rather than byte-identical execution of its optimizer or the published scattershot algorithm [8].

Table 15. Component fitting assumptions. The onset lower bounds are absolute component-index bounds; they do not enforce pairwise onset separation after fitting.

Setting	Implemented value or rule
Candidate count	One to four, subject to feasible onset bounds
Component duration	0.100 to 1.000 s
Lateral / forward displacement	[-5, 5] / [0, 20] tracker units
Onset lower bounds	Component index times 0.050 s (index begins at zero)
Onset upper bound	max(movement duration - 0.050 s, 0.050 s)
Matched fitting budget	Two restarts; at most 400 function evaluations per restart
Primary order choice	Smallest count with $E \leq 0.05$; otherwise smallest with $E < 0.10$; otherwise minimum E

An alternative count minimized the stored BIC-style criterion below, where N was three times the padded number of time points and there were four fitted parameters per component. The criterion was retained as a model-order sensitivity; its residual independence assumptions are not established for filtered velocities.

$$\text{BIC}_K = M \log(\text{RSS}_K/N) + 4K \log N.$$

[Jason: Are this pulse basis, displacement bounds, 100-ms minimum duration, endpoint padding and order-selection rule appropriate for these interception movements, or should the decomposition follow a different lab convention?]

10.2 Matched processing of recorded and generated inputs

The original decomposition comparison used different numerical budgets and input representations for recorded and generated movements. A matched evaluation therefore started from the same type of model-target window: 100 planar positions from target motion to the final sample. Both inputs underwent the same timing-based crop to 100 movement points, restoration of physical sampling rate as 99 divided by movement duration, low-pass filtering and component fitter.

The recorded input used measured timing and retained its earlier acquisition preprocessing; generated input used predicted timing. Matching the downstream operator does not remove these upstream differences. The resulting endpoint describes the model's filtered representation, rather than unprocessed biological components.

All 2,376 recorded query windows and 1,680 CVAE-generated windows were fitted: ten generated trials per participant, capacity and neural seed at $n=3$ and $n=8$. The VAE $n=8$ follow-up added 840 generated windows. Recorded input arrays and durations were verified before their existing matched fits were reused. Every fitting attempt completed; numerical convergence was reported separately and did not determine inclusion.

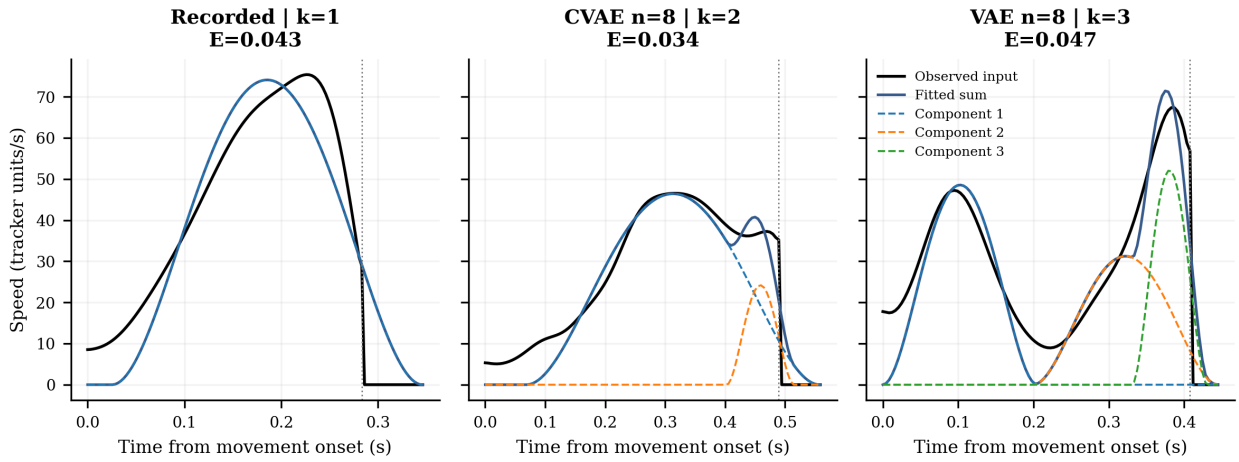


Figure 15. Saved minimum-jerk fits for subject01: the first recorded query job and first seed-42 generated job in identifier order for CVAE $n=8$ and VAE $n=8$. The black curve is the input speed (recorded or generated), blue is the norm of the fitted vector sum, and dashed curves are component speeds. Dashed magnitudes need not add to blue. The vertical dotted line marks the movement endpoint; later input speed is padded with zeros. No optimizer was rerun to choose these illustrations.

10.3 Component-distribution endpoints

Five continuous descriptors were compared using KS: normalized fit error; first-component duration; first-component displacement amplitude; the fraction of total component amplitude assigned to later components; and mean overlap percentage. Components were sorted by fitted onset. Amplitude was the norm of the displacement vector, not peak speed. A one-component fit had zero secondary-amplitude fraction and zero mean overlap.

For adjacent components, overlap was the shared interval from the second onset to the first end divided by the interval from the first onset to the second end, multiplied by 100 and clipped to $[0,100]$. The implementation then averaged adjacent-pair values. This operational definition was retained for both recorded and generated fits.

Selected-count distributions over one to four components were compared by total variation and Jensen-Shannon divergence (base-two logarithms). Total variation sums the absolute differences in category proportions and divides by two. JSD averages each distribution's divergence from their equal-weight mixture. BIC-selected counts provided two additional sensitivity endpoints.

$$\text{TV}(p, q) = \frac{1}{2} \sum_{k=1}^4 |p_k - q_k|.$$

10.4 Fidelity and sensitivity results

Table 16. Participant-balanced matched component discrepancies. The reference is the 95th percentile of an empirical resampling distribution, not a population confidence interval or formal acceptance threshold.

Discrepancy	CVAE n=3	CVAE n=8	VAE n=8	Reference 95th pct.
Fit error KS	0.339	0.308	0.328	0.282
First duration KS	0.455	0.409	0.406	0.283
First amplitude KS	0.377	0.351	0.362	0.283
Secondary amplitude fraction KS	0.287	0.286	0.302	0.212
Overlap KS	0.267	0.268	0.279	0.209
Count total variation	0.214	0.232	0.236	0.163
Count JSD	0.076	0.087	0.089	0.054
BIC-count total variation	0.179	0.158	0.169	-
BIC-count JSD	0.059	0.043	0.053	-

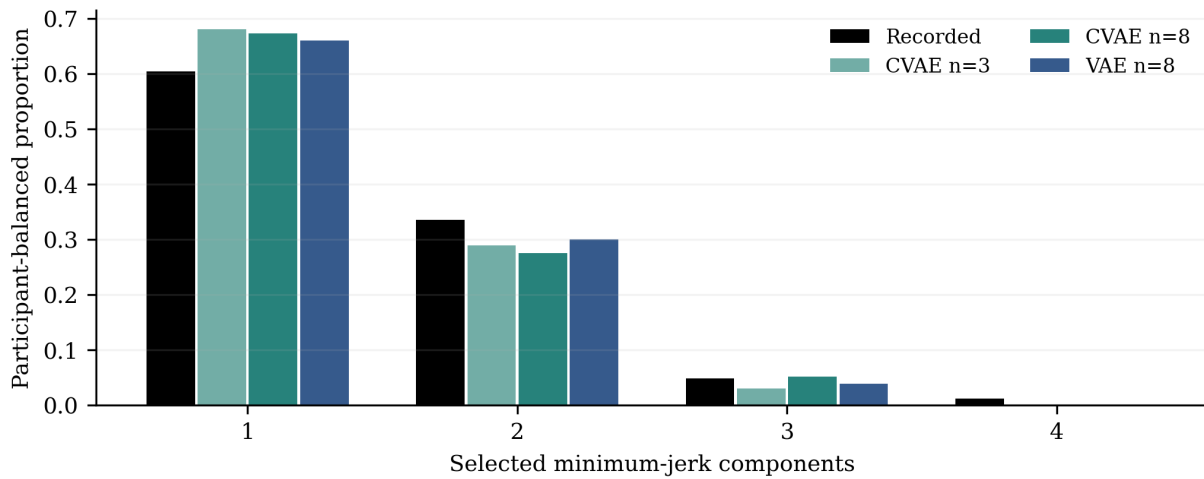


Figure 16. Matched selected-count proportions, averaging within participants and then equally across participants, with seed averaging for generated movements. Aggregate similarity can coexist with participant-specific discrepancies; Table 16 computes distances within participants first.

Matching the procedure reduced the historical CVAE count discrepancy (approximately 0.353/0.355 at $n=3/8$) to 0.214/0.232. The remaining mismatch was not eliminated. Each of the seven principal discrepancies exceeded its empirical reference's 95th percentile for both CVAE capacities and VAE $n=8$. The reference drew independent resamples from each participant's recorded query distribution at the observed query size and a generated sample size of ten. Five hundred replicates each averaged three seed-like draws and then participants. It accounts for this finite-sample discrepancy conditional on the empirical records, not fitting, training or population uncertainty.

None of nine paired $n=3$ versus $n=8$ CVAE component comparisons survived BH correction. VAE $n=8$ did not demonstrate an improvement over CVAE $n=8$ on these endpoints either: all nine adjusted p-values were 0.8643. Better 11-feature generation distances therefore did not establish better minimum-jerk component fidelity.

10.5 Optimizer convergence and alternative assumptions

All 4,056 selected fits in the matched recorded/CVAE comparison converged, although 505 fits had at least one nonconverged alternative candidate order. For VAE $n=8$, 839 of 840 selected fits converged and 734 had every candidate order converge. The reported comparison retained the declared error-based selection without excluding nonconverged candidates.

Table 17. Fixed higher-budget checks using four restarts and at most 700 evaluations. Selection used the first two query trials or first two seed-42 generated samples per participant, rather than selecting difficult fits after viewing their outcomes.

Subset	Higher-budget fits	Same selected count	Selected fits converged
CVAE generated	112	111/112	112/112
Recorded	56	56/56	56/56
VAE $n=8$ generated	56	56/56	56/56

One additional VAE-generated fit whose selected candidate had not converged was separately rerun with the higher budget. It converged with the same two-component count and nearly unchanged normalized error (0.03781723 to 0.03781690). This was a targeted diagnostic, separate from the fixed subset; the original fixed-budget score remained in the reported comparison.

Earlier assumption sensitivities used two randomly selected recorded trials per participant (56 total, fixed seed 20260824). Changing the filter cutoff to 5 Hz retained the error-selected count in 29/56 cases (51.8%). Changing both the minimum duration and indexed onset lower-bound step to 167 ms retained it in 31/56 (55.4%). The corresponding BIC-count agreement was 34/56 and 1/56. These historical runs used the earlier processing scheme, so they should not be combined with the matched results as a factorial experiment under one common pipeline.

[Jason: Which aspects of the decomposition are scientifically meaningful to compare here: count, amplitude, overlap, fit error, or another descriptor? Should minimum-jerk fidelity remain a secondary diagnostic in this project?]

11 Event timing and retained early movement

The interpretation of initiation time depends on the event chosen as time zero. The current model begins at target motion, while marker 5 denotes target appearance. In the available recordings, the appearance-to-motion interval ranged from 0.20 to 0.50 s, with median 0.35 s. Movement during this earlier interval remains in the recordings but is outside the modeled window.

A dedicated event audit examined all 4,732 retained trials without removing or relabeling them. There were 357 zero-initiation labels (7.5%). In every such trial, filtered 3D speed already exceeded the onset threshold for the required three frames at target motion; none of these labels resulted from a failed onset search. Filtered planar speed showed at least 50 ms of continuous pre-motion threshold exceedance in 109 of those trials, and raw planar speed did so in 100.

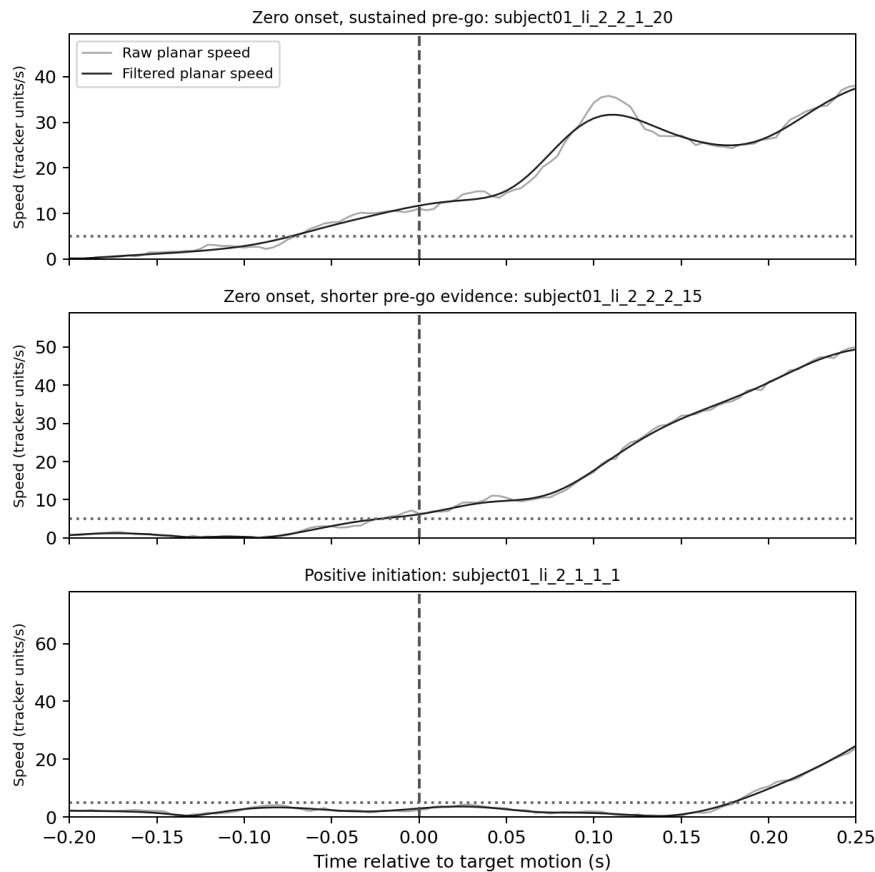


Figure 17. Recorded raw and filtered planar speed around target-motion onset. The three examples were selected by proximity to the median pre-motion threshold duration within their stated groups. Dashed vertical lines mark target motion; horizontal dotted lines mark the speed threshold. The modeled onset rule uses 3D speed, so planar plots illustrate rather than exactly reproduce that rule.

Early movement was retained as behavior of interest. Zero-phase filtering can spread a change backward in time, and raw and filtered checks differ. The audit therefore describes threshold behavior without assigning its cause. The 357 zero labels contributed 74.6% of the centered squared variation in $\log(\text{initiation} + 0.001)$; that statistic is target variation, not residual loss or a measured explanation for the timing errors.

[Jason: Should the modeled interval begin at target appearance to include evidence gathering or movement before target motion? Please confirm the event alignment and whether the final tracker sample is the intended interception endpoint. A change in the modeling window would require a matched preprocessing and model reevaluation.]

12 Interactive dashboard as the project deliverable

The model comparisons were translated into an interactive dashboard for exploring learned movement representations, generated trajectories, predicted durations and feature distributions. The interface exposes the tested families so that the choice of model can follow the purpose of an exploration. It uses frozen reference models and precomputed evaluation artifacts; moving a slider does not train a new model.

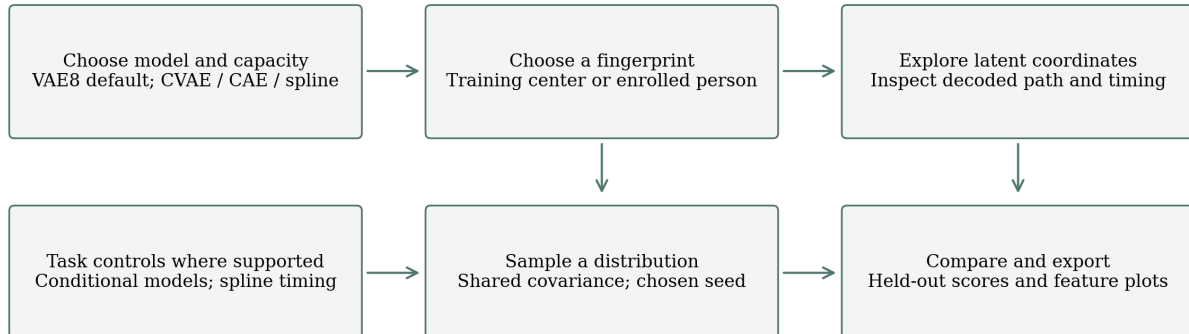


Figure 18. Main dashboard controls and outputs. A latent coordinate change produces a model-dependent movement; distribution sampling adds the shared training-derived variability around the selected center.

Table 18. Supported dashboard references. The condition-only Ridge remains an evaluation baseline; it has no personal latent fingerprint and is not a live reference model.

Reference family	Available dimensions	Use of task controls
CVAE	2, 3, 4, 8	Encoder and decoder are conditional
CAE	3, 8	Encoder and decoder are conditional
VAE	3, 8	Inactive; outputs model the condition mixture
Spline + PCA	2, 3, 4, 8	Timing regression only; shape depends on PCA scores

The default is VAE $n=8$ because it had the lowest cohort-average KS, energy and MMD-squared generation discrepancies in the tested matrix. The choice is endpoint-specific and based on the evaluated cohort, with no separate post-selection validation. CVAE $n=3$ remains available for the three-number, explicitly conditional setting; spline and CAE provide useful alternatives for reconstruction. No $n=16$ result is available from the matched study.

Live references use fold 0 and neural seed 42. The user can choose the training-population center or one of that fold's enrolled test-participant context centroids, then move coordinates in units of training-code standard deviation. The distribution panel samples 25 to 500 latent points, default 100, using a user-selected random seed and the training covariance. Conditional generation uses the selected task controls; the unconditional model ignores them.

The held-out validation section instead displays the exported evaluations across all four folds at seed 42, using 120 generated samples per participant. Summary score cards use the study's available training seeds. These displays answer different questions and are labeled accordingly. A live center-decoded curve should not be presented as a recorded trial or as a quantitative validation result.

12.1 Scope of the deliverable

The dashboard includes trajectory, speed and illustrative minimum-jerk views, feature distributions, held-out comparisons, latent associations, study diagnostics and report downloads. Its live decomposition uses one restart and at most 300 evaluations; this exploratory fit does not replace the matched offline component evaluation. Successful software checks establish that the selected model and assets load and execute; they do not validate an interpretation of the sliders as cognitive strategies.

[Jason: For the next stage, is the priority a strict three-number representation, the best observed marginal-distribution fidelity, or reliable control by experimental condition? The current results support different model choices for these aims.]

12.2 Additional evaluated capacities and evidence boundaries

Table 19. Additional capacities in the original CVAE/spline sweep. These complement the n=3 and n=8 matched comparisons; CAE and VAE were not evaluated at n=2 or n=4.

Model	n	Trajectory MSE	Initiation MAE (ms)	Movement MAE (ms)	Mean KS	Energy	MMD squared
Spline + PCA	2	0.1970	56.56	50.89	0.355	1.141	0.214
CVAE	2	0.3247	39.82	46.78	0.310	0.891	0.163
Spline + PCA	4	0.0765	56.24	51.21	0.302	0.899	0.167
CVAE	4	0.1018	35.70	44.84	0.263	0.668	0.113

Condition-only population generation

The condition-only Ridge baseline also generated stochastic movements without a personal code. Separate Ridge models predicted coordinates and standardized log-durations from the five task values, with validation-selected alpha from 0, 0.01, 0.1, 1, 10 and 100. Generation added a paired trajectory/timing residual from a training trial in the same start-category and side stratum, falling back to the full training pool if needed. Continuous speed entered the regression rather than requiring an exact residual match.

Its cohort-average generation discrepancies were mean KS 0.277264, energy 0.818722 and MMD squared 0.125854. These competitive marginal scores, despite much poorer trajectory reconstruction, further show why generation and compression require separate evaluation. The baseline has no participant centroid; it cannot itself demonstrate that a personal fingerprint contributes information.

The results form a sequence of distinct tests: compression of observed paths; generation of query-feature distributions from personal context; replacement of that context to test its contribution; prediction of timing and behavioural summaries; and secondary checks of conditioning, component fitting and event semantics. An advantage in one test does not establish an advantage in the others.

The audit reproduced the saved neural outputs, checked condition routing and participant separation, and corrected confirmed implementation defects in auxiliary code. All 96 frozen neural checkpoint hashes remained unchanged. The model matrix used four declared participant folds and three neural seeds; it did not enumerate all possible participant splits or test every architecture and capacity. No controlled learning-curve experiment was performed to attribute a limitation to participant count.

The original 56-test benchmark remains in the evidence archive. This draft uses the expanded exploratory all-pairs comparison (140 tests). Separate BH families were used for that comparison, CVAE fingerprint replacements (12), VAE fingerprint replacements (6), timing-head sensitivities (8), condition-feature comparisons (44), CVAE component-capacity contrasts (9) and VAE-CVAE component contrasts (9). Holm sensitivity over the expanded 140 and the two condition-trajectory contrasts is stated where relevant. These adjustments were not a single correction over every exploratory analysis.

Training folds overlap. The historical character-sum participant seed scheme also maps multiple participant identifiers to the same sampling seed, creating shared simulation randomness. Neither multiple-testing adjustment nor repeated training seeds removes those dependence limits. They should remain explicit when interpreting the strength and scope of the evidence.

This detailed version assembles the methods, results and domain questions for laboratory review. The final introduction, abstract and discussion will be written around the agreed scientific objective and the results retained with the advisors. The eventual eight-page main paper can point to the complete supplementary evidence without presenting every diagnostic as a separate primary finding.

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